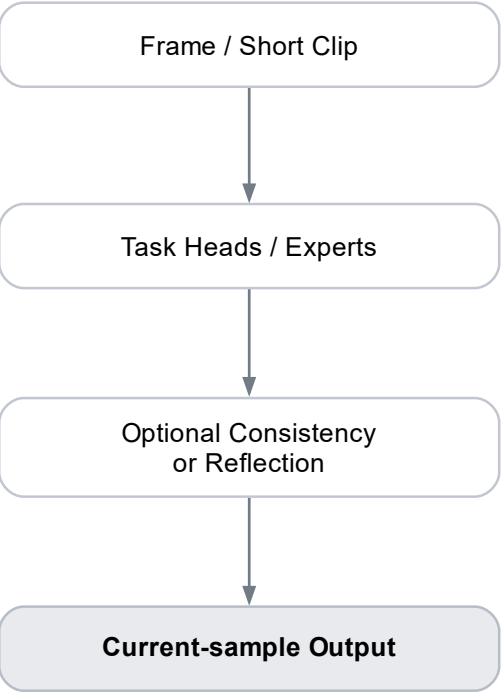


From Clip-level Recognition to Benefit-Routed Long-Video Reasoning

Why stateful and selective verification is needed for causal surgical video understanding

PREVIOUS METHODS

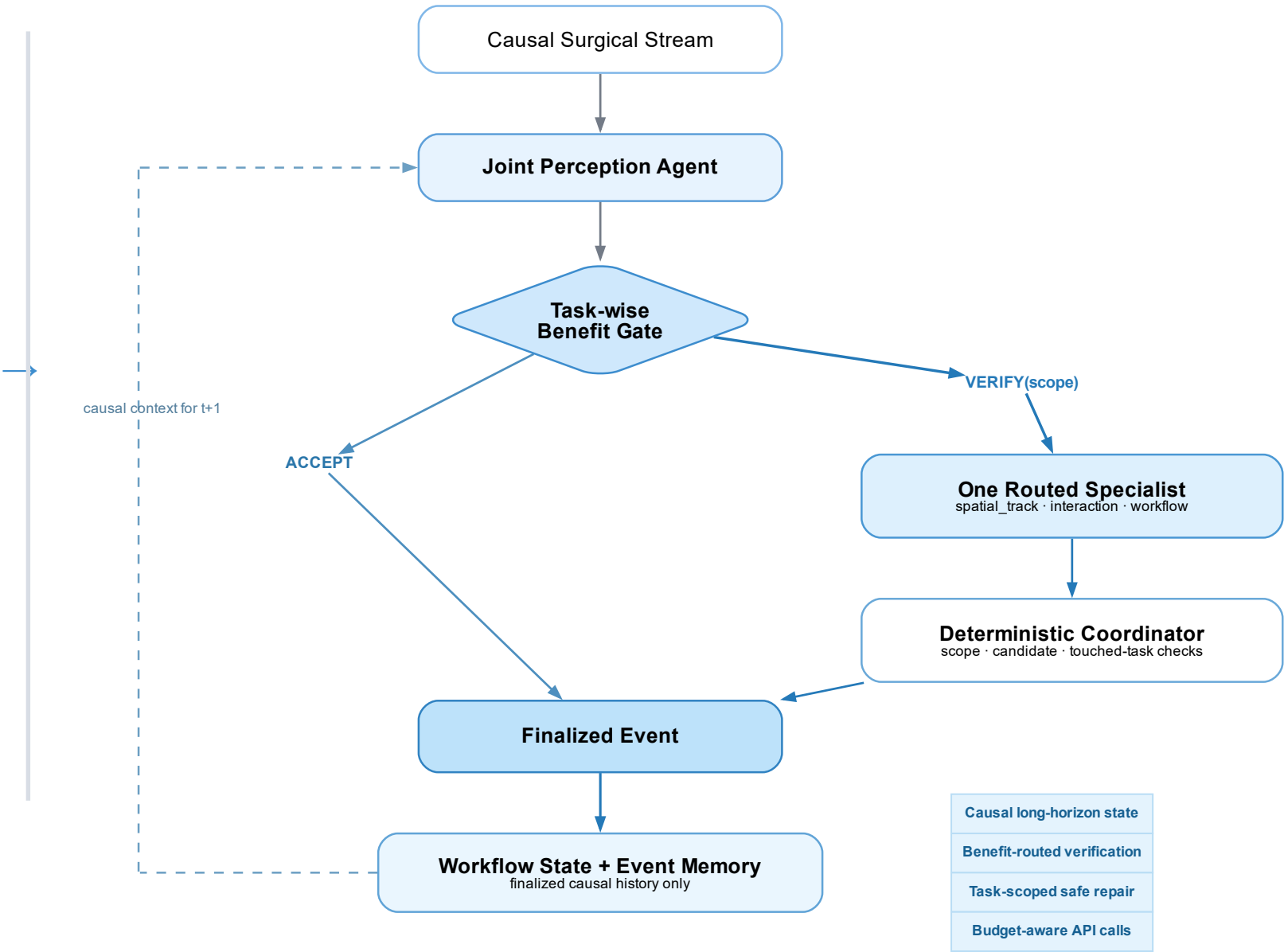
Sample-wise clip understanding



Local temporal context
Independent sample processing
No explicit persistent video state

OURS

Stateful long-video surgical agents



Causal long-horizon state
Benefit-routed verification
Task-scoped safe repair
Budget-aware API calls